
Analysis of the Use of Liquid Hydrogen for Storage in the Road Haulage Sector

EN426: Electricity Generation and Hydrogen



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1 Introduction

Throughout history, humanity has relied on energy to survive and thrive. First came fire, then muscle power, and finally natural energy sources such as wind, coal, oil, and electricity. All of these energy sources have enabled humans to develop and shape the world in their own way. Present in every sector, energy is indispensable to our society. But not all energy sources are created equal.

Some have a very high environmental cost: they pollute. Others do not. Unfortunately, today, nearly 87% of the energy used worldwide comes from fossil fuels, which pollute. This proportion is even higher in the trucking industry, where nearly 95% of trucks run on diesel. Although this method offers great range, it releases enormous amounts of carbon dioxide (CO₂), nitrogen oxides (NO_x), and fine particulate matter into the atmosphere—highly polluting gases that exacerbate the greenhouse effect and thus contribute to global warming. People who care about the environment

want to find a stable and credible alternative to these polluting fossil fuels in order to replace them with greener energy sources. In the road transport sector, hydrogen is seen as a promising alternative to diesel. Indeed, the use of hydrogen produces no CO emissions or fine particulate matter, but only releases water. In addition to being low-carbon, hydrogen can be used in two different ways. Either as fuel in an internal combustion engine, where it emits no CO but only NO_x; or in a fuel cell, where it produces electricity through an electrochemical reaction, which is then used to power the truck. However, hydrogen has one problem: storage. Storing hydrogen is more complicated than storing diesel. In fact, to take up as little space as possible and thus increase the amount of available energy, hydrogen must either be compressed or stored in liquid form. However, the storage pressure for hydrogen is very high (approximately 350 to 700 bar), which requires ultra-resistant tanks. And liquid hydrogen requires a very low temperature of -253 °C, which also complicates its storage. Our work has therefore focused on hydrogen storage, whether at service stations or inside the trucks themselves, to determine the best way to store hydrogen.

2 Hydrogen Technologies in Heavy-Duty Transport

2.1 The Hydrogen Internal Combustion Engine (H-ICE)

The hydrogen internal combustion engine (H-ICE) adopts the same architecture as a conventional petrol or diesel engine. Hydrogen is injected into the cylinders, mixed with air, and then ignited. The combustion generates pressure that drives the pistons and produces mechanical work. This technology offers several notable advantages for the heavy transport sector. Its implementation represents a straightforward technological transition, allowing manufacturers to reuse existing production lines and established engineering expertise. The engine demonstrates high robustness, making it well-suited to demanding environments involving dust, vibrations, and extreme temperatures. Its cost is significantly lower than that of a fuel cell system, and its maintenance requirements are familiar to road transport operators. However, this approach also presents inherent limitations. Its thermal efficiency remains moderate, ranging from 25 to 40%, which is comparable to petrol engines. The high combustion temperatures produce nitrogen oxides (NO_x), requiring an additional depollution system. Furthermore, efficiency is reduced in urban or stop-and-go driving conditions.

2.2 The Fuel Cell

The fuel cell operates through an electrochemical reaction rather than combustion. Hydrogen is dissociated into protons and electrons; the electrons flow through an external circuit to generate

electricity, while oxygen from the air combines with the protons to form water. This electricity then powers an electric motor that drives the wheels. Fuel cell systems offer a high efficiency rate, and produce zero pollutant emissions, only water vapour. Their silent operation makes them particularly suited for urban environments and regular duty cycles. On the other hand, fuel cells involve a high manufacturing cost due to noble materials and membrane components. Their operational lifespan is more limited under the intensive duty cycles typical of heavy freight. Additionally, they are sensitive to impurities in the hydrogen supply and to thermal variations, and require complex thermal and humidity management systems.

3 Focus on the Hydrogen Internal Combustion Engine

3.1 Thermodynamic Principles

The hydrogen internal combustion engine operates according to the same thermodynamic cycles as petrol engines: the Otto cycle for spark ignition, and a modified Diesel cycle for compression ignition. Hydrogen exhibits a number of distinctive properties: an extremely wide flammability range, a high flame speed, a very low activation energy, and the complete absence of carbon atoms, meaning no CO is produced. These characteristics significantly influence engine design. They require fine-tuned injection management, precise temperature control, and prevention of two specific phenomena: knocking caused by premature auto-ignition and flashback, where the flame propagates back into the intake manifold due to hydrogen's low ignition energy.

3.2 Thermal Management: Controlling Combustion Temperature

Although hydrogen has a higher adiabatic flame temperature than fossil fuels (approximately 2,300–2,400 °C compared to 2,100–2,200 °C for petrol or diesel), the actual temperature reached inside the combustion chamber of a modern H engine is significantly lower than this theoretical value. Manufacturers exploit hydrogen's particular properties and apply several complementary strategies to maintain thermal loads within acceptable limits.

3.2.1 Lean-Mixture Operation

Hydrogen has an exceptionally wide flammability range from 4% to 75% by volume in air which allows the engine to operate with a very large excess of air, well beyond what is achievable with petrol. Running in lean-mixture mode produces several beneficial effects: it reduces combustion temperature by up to 200 °C below stoichiometric conditions, significantly lowers NO formation, and reduces thermal stress on cylinder walls and pistons. This strategy is the primary approach adopted by manufacturers to ensure engine durability.

3.2.2 Injection Strategy

Hydrogen injection can be either indirect (into the intake manifold) or direct (into the combustion chamber). Direct injection, which is increasingly favoured in modern H engines, provides more precise control over the air-fuel mixture distribution. It avoids locally rich zones that would otherwise generate excessive temperatures, enables tighter stoichiometric control, and reduces the risk of flashback. This precise management of fuel distribution contributes to limiting thermal peaks within the chamber.

3.2.3 Compression Ratio and Knock Prevention

Due to hydrogen's very low activation energy and high flame speed, the risk of auto-ignition, knocking, and pre-ignition is elevated. To mitigate these phenomena, H engines operate with a reduced compression ratio, use advanced ignition timing management, and employ enhanced cooling systems. These measures also indirectly limit the maximum temperature reached in the combustion chamber.

3.3 Performance and Efficiency

The overall efficiency of the H internal combustion engine remains constrained by the inherent thermodynamic losses of the combustion process. However, hydrogen's unique properties enable several optimisations: lean-mixture operation reduces pumping losses, the high flame speed enables faster, more complete combustion, and the absence of carbon eliminates soot formation entirely. In practice, H-ICE trucks currently achieve thermal efficiencies in the range of 30 to 40%, which represents a notable improvement over diesel engines in certain operating conditions, while remaining below the efficiency of fuel cell systems.

3.4 Component Design and Adaptation

Even with these temperature control strategies, certain engine components must be specifically adapted to withstand the particular constraints imposed by hydrogen combustion. Pistons are equipped with undercrown oil-jet cooling systems and manufactured from reinforced aluminium-silicon alloys. Their geometry is optimised to minimise hot spots. Intake and exhaust valves, as well as their seats, are made from high-temperature alloys such as Inconel or nickel-based materials, and are reinforced to withstand hydrogen's high flame speed. The cylinder head features optimised cooling circuits and enhanced thermal management around the ignition zones. It is worth noting that many of these technical solutions are already well-established in modern turbocharged petrol engines, which significantly simplifies their adaptation for hydrogen applications.

4 Station-side liquid hydrogen storage and boil-off management

This section is intended to be inserted before the current section on truck storage. Before analysing how liquid hydrogen behaves inside the truck tank, it is necessary to understand what happens at the refuelling-station level. A station tank is larger, remains filled for longer periods, and is connected to pipes, valves, vents and dispensers. These elements create additional thermal bridges and safety constraints.

The objective is therefore not to repeat the complete truck-tank model, but to use the same physical logic: heat enters the cryogenic system through insulation, the liquid reaches its boiling temperature, and the incoming heat is then converted into vaporisation. The key question becomes economic as much as thermal:

If a small quantity of liquid hydrogen evaporates every day, is it worth recovering this boil-off gas, compressing it, or reliquefying it? Or is controlled venting more rational for a well-insulated tank?

The answer depends on three parameters: the real insulation performance, the market value of hydrogen, and the cost of the equipment required to recover the gas safely.

4.1 Thermal insulation principle

Liquid hydrogen is stored close to 20.3 K, corresponding to approximately -253°C . This produces a very large temperature difference with the ambient environment. Even with a high-quality cryogenic tank, heat transfer can only be reduced, not completely eliminated.

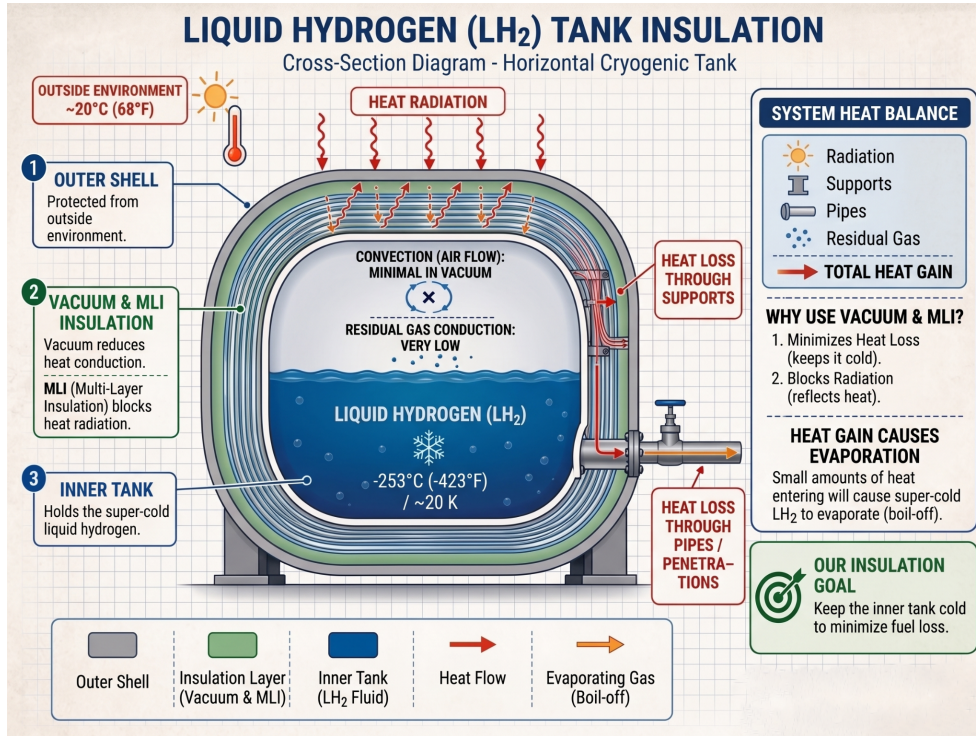


Figure 1: Cross-section principle of a liquid hydrogen tank: outer shell, vacuum space, MLI insulation, inner tank, radiation, supports and pipe penetrations.

The insulation system is based on two main barriers. First, the vacuum gap suppresses most convection and reduces gas conduction. Second, the multi-layer insulation (MLI) reduces radiative heat transfer by reflecting thermal radiation between many thin layers. However, some heat still enters through residual gas, mechanical supports, pipes, valves and instrumentation penetrations.

The total heat input can therefore be written as:

$$\dot{Q}_{\text{tot}} = \dot{Q}_{\text{rad}} + \dot{Q}_{\text{gas}} + \dot{Q}_{\text{sup}} + \dot{Q}_{\text{pipes}} + \dot{Q}_{\text{op}} \quad (1)$$

where $\dot{Q}_{\text{op}}$ represents additional operational losses during filling, dispensing, pressure regulation and valve operation.

The radiative contribution is estimated by:

$$\dot{Q}_{\text{rad}} = \varepsilon_{\text{eff}} \sigma A (T_{\text{amb}}^4 - T_{\text{LH}_2}^4) \quad (2)$$

where ε_{eff} is the effective emissivity of the MLI system, σ is the Stefan-Boltzmann constant, and A is the effective heat-transfer surface.

The residual-gas conduction term may be approximated as:

$$\dot{Q}_{\text{gas}} = \frac{k_{\text{eff}} A}{e_{\text{gap}}} (T_{\text{amb}} - T_{\text{LH}_2}) \quad (3)$$

and the solid conduction through supports and pipes can be grouped as:

$$\dot{Q}_{\text{sup}} = \sum_i \frac{k_i A_i}{L_i} (T_{\text{amb}} - T_{\text{LH2}}), \quad \dot{Q}_{\text{pipes}} = \sum_j U_j A_j (T_{\text{amb}} - T_{\text{LH2}}) \quad (4)$$

For the economic discussion, it is more convenient to group all these effects into one equivalent thermal flux:

$$q''_{\text{eff}} = \frac{\dot{Q}_{\text{tot}}}{A} \quad (5)$$

This parameter directly represents the real quality of the tank insulation. A low value means that radiation, residual gas conduction, supports and pipe penetrations are well controlled. A higher value means degraded vacuum, excessive thermal bridges or unfavourable operation.

4.2 Boil-off calculation

Once the liquid hydrogen reaches its boiling point, incoming heat no longer raises the liquid temperature significantly. It is mainly used as latent heat to vaporise the liquid. The boil-off mass flow rate is therefore:

$$\dot{m}_{\text{BOG}} = \frac{\dot{Q}_{\text{tot}}}{L_v} \quad (6)$$

For a full day:

$$m_{\text{BOG,day}} = \frac{q''_{\text{eff}} A \times 86400}{L_v} \quad (7)$$

The station case is based on the following general assumptions. The objective is not to model a specific commercial station, but to use realistic orders of magnitude for a liquid hydrogen storage tank.

Table 1: Reference assumptions for the station tank.

Parameter	Value used	Comment
Stored LH ₂ mass	3500 kg	Typical order of magnitude of one LH ₂ tanker delivery
LH ₂ density	70.8 kg m ⁻³	Approximate value near 20 K
Equivalent volume	49.4 m ³	$V = m/\rho$
Effective heat-transfer surface	82 m ²	Horizontal cryogenic station tank assumption
Latent heat of vaporisation L_v	445 kJ kg ⁻¹	Energy required to evaporate 1 kg of LH ₂
Industrial electricity price	0.19 €/kWh	EU non-household order of magnitude

The resulting boil-off rate is shown in Table 2. The important point is that the losses are controlled mainly by q''_{eff} , not only by the size of the tank.

Table 2: Daily boil-off as a function of insulation quality.

Insulation condition	q''_{eff} [W/m ²]	BOG [kg/day]	Mass loss [%/day]
Excellent vacuum + MLI	0.15	2.39	0.068
Base case: good vacuum + MLI	0.30	4.78	0.137
Important supports and penetrations	0.60	9.55	0.273
Degraded vacuum / unfavourable operation	1.20	19.10	0.546

A well-insulated tank does not eliminate boil-off; it makes the boil-off small. This distinction is essential for the economic conclusion: a small boil-off flow has value, but it may not be large enough to justify a dedicated recovery system.

4.3 Economic value of the evaporated hydrogen

The first economic step is to estimate the value of the hydrogen that would be lost if the BOG were vented. Three hydrogen prices are considered:

$$P_{\text{H}_2} = 8, 12, 16 \text{ €/kg} \quad (8)$$

These values represent a sensitivity range: about 8 €/kg corresponds to an aggressive high-utilisation truck-refuelling target, while 12–16 €/kg represents more conservative delivered-hydrogen prices.

The daily value of vented hydrogen is:

$$C_{\text{vent}} = m_{\text{BOG,day}} P_{\text{H}_2} \quad (9)$$

If a recovery system is installed, it will not recover 100% of the gas in real operation. A practical recovery factor of 90% is assumed:

$$C_{\text{rec,gross}} = 0.90 m_{\text{BOG,day}} P_{\text{H}_2} \quad (10)$$

Figure 2 compares the gross recoverable value of BOG with two indicative daily fixed costs: one for a small BOG compression system and one for a small reliquefaction system. These fixed costs represent amortisation, maintenance, instrumentation, hydrogen sensors, safety devices, inspections and system availability. They are deliberately approximate, because the exact value depends strongly on the supplier and station design.

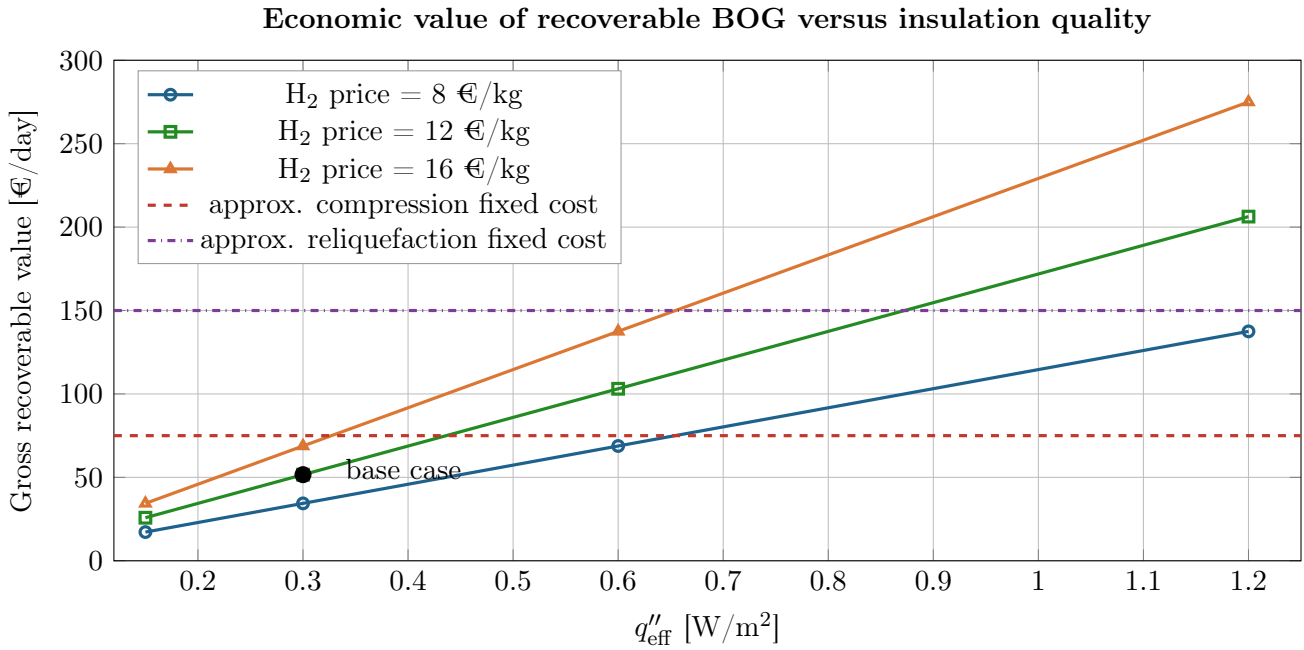


Figure 2: With good insulation, the value of recoverable boil-off remains below or close to the fixed cost of compression, and far below the fixed cost of reliquefaction. Recovery becomes economically relevant only if insulation is degraded, BOG flow increases, or hydrogen price is high.

4.4 Compression or reliquefaction of the BOG

Three possible strategies can be considered:

1. **Controlled venting:** the gas is released through a safe vent stack. Hydrogen is lost, but the system remains simple.
2. **Compression:** the BOG is compressed to a gaseous buffer or to a pressure level compatible with vehicle refuelling.
3. **Reliquefaction:** the BOG is cooled again to cryogenic temperature and returned to the liquid tank.

The electricity cost per kilogram of treated gas is:

$$C_{\text{elec}} = E_{\text{sp}} P_{\text{elec}} \quad (11)$$

where E_{sp} is the specific energy consumption and P_{elec} is the electricity price. The assumptions used are given in Table 3.

Table 3: Treatment assumptions for BOG recovery.

Option	Specific energy [kWh/kg]	Electricity cost [€/kg]	Daily fixed cost used
Compression to 350 bar	2.0	0.38	75 €/day
Compression to 700 bar	4.0	0.76	75 €/day
Reliquefaction	11.0	2.09	150 €/day

The net saving compared with venting is:

$$S_{\text{net}} = \eta_{\text{rec}} m_{\text{BOG,day}} P_{\text{H}_2} - \eta_{\text{rec}} m_{\text{BOG,day}} E_{\text{sp}} P_{\text{elec}} - C_{\text{fixed}} \quad (12)$$

with $\eta_{\text{rec}} = 0.90$. If $S_{\text{net}} > 0$, recovery is economically better than venting. If $S_{\text{net}} < 0$, the recovery system costs more than the hydrogen it saves.

Figure 3 applies this equation to the base case of a well-insulated tank: $q''_{\text{eff}} = 0.30 \text{ W/m}^2$, corresponding to approximately 4.78 kg/day of BOG.

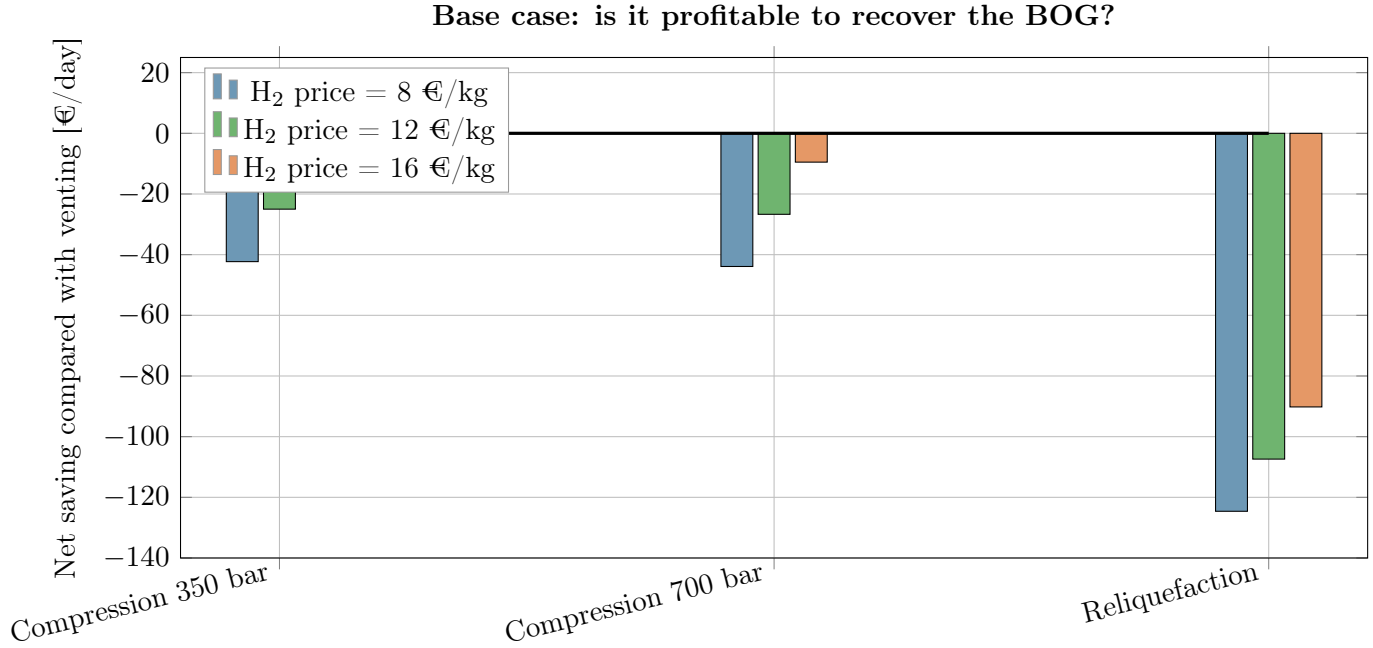


Figure 3: For the base case, every value is negative: the tank is sufficiently well insulated that the BOG flow is too small to justify a dedicated compression or reliquefaction system. Compression gets close to break-even only at a high hydrogen price, whereas reliquefaction remains clearly unattractive.

4.5 Safety constraints if BOG is not managed correctly

The previous result does not mean that BOG can simply be ignored. If the gas is not vented, compressed or otherwise managed, pressure inside the tank increases. This may activate relief valves or lead to uncontrolled discharge. In addition, hydrogen has a very wide flammability range in air, approximately 4–75% by volume, and a very low ignition energy.

Hydrogen does not explode spontaneously. The risk appears when three elements occur together: hydrogen, air, and an ignition source. A poorly managed BOG release can create a flammable cloud, especially under roofs, inside enclosures, around parked vehicles, or in poorly ventilated areas.

The combustion condition can be represented simply as:



Therefore, a station handling LH₂ must include at least:

- vertical venting at height, away from people, vehicles and air intakes;

- hydrogen detectors in critical zones;
- natural or forced ventilation to prevent accumulation;
- pressure relief valves and rupture discs;
- ATEX zoning around likely leak points;
- electrical grounding to reduce electrostatic ignition risk;
- inert-gas purging during maintenance;
- continuous monitoring of tank pressure and vacuum quality.

Safety equipment is one reason why BOG recovery is not only an electricity-cost problem. A compressor or reliquefier connected to boil-off gas must also include control valves, buffers, sensors, shutdown logic, purging systems and certified equipment. For a very small BOG flow, these fixed costs dominate the economic balance.

4.6 Final interpretation

The calculations show that the reference tank is well insulated. In the base case, the station produces about 4.78 kg/day of BOG, which corresponds to only 0.137% of the stored mass per day. This loss is technically real, but economically too small to justify a dedicated recovery system.

For the base case, if hydrogen is valued at 12 €/kg, the gross daily value of the recoverable BOG is about 52 €/day. This is lower than the assumed fixed cost of a small compression system, and much lower than the fixed cost of a reliquefaction system. Once electricity is included, the net saving remains negative.

The conclusion is therefore clear for this type of well-insulated tank:

- **reliquefaction is not profitable**, because its electrical consumption and fixed cost are too high for such a small mass flow;
- **compression is technically possible but not economically attractive** in the base case, because the BOG flow is too low to amortise the equipment;
- **controlled venting or integration into an already existing gaseous-hydrogen line** is more reasonable for small residual BOG flows;
- **the priority should be insulation quality**: good vacuum, efficient MLI, limited pipe penetrations, reduced thermal bridges and good operational planning.

This result is important for the rest of the report. Liquid hydrogen is attractive because of its high volumetric density, but its station and vehicle use must be judged together with thermal losses, safety systems and operating costs. A well-insulated tank reduces boil-off so effectively that recovering the residual gas may become less profitable than safely managing a small controlled loss.

5 Hydrogen Storage for Heavy-Duty Trucks

5.1 Compressed Gaseous Hydrogen (CGH)

Current hydrogen trucks store compressed hydrogen (CGH) at pressures of 350 or 700 bar in carbon fibre composite tanks. This technology is mature and commercially available, but its volumetric energy density is inherently limited approximately 40 g/L at 700 bar. This results in heavy, bulky storage systems, and restricts the operational range of heavy-duty trucks to approximately 400 to 600 kilometres, which is not enough for long travel.

5.2 Liquid Hydrogen (LH), The Case of Cryogenic Storage

Liquid hydrogen (LH), stored at 253 °C (20 K), achieves a volumetric density of 71 g/L — nearly twice that of compressed gas at 700 bar. For long-distance trucks requiring a range exceeding 800 kilometres, this represents a decisive advantage: smaller tank volumes, more space for cargo, and faster mass-flow refuelling. These factors make LH storage particularly relevant for high-capacity logistics applications.

5.3 The Liquefaction Process

Producing liquid hydrogen requires a four-stage industrial process. The first stage is purification. Hydrogen must be of ultra-high purity because any impurity water vapour, nitrogen freezes instantly at 20 K and blocks the cryogenic circuits. The second stage involves compression and pre-cooling. The gas is compressed to between 20 and 80 bar, then pre-cooled to 196 °C using a liquid nitrogen refrigeration cycle before further cooling steps are applied. The third stage often overlooked but technically critical is the ortho-to-para hydrogen conversion. Hydrogen exists in two nuclear spin states: ortho-hydrogen and para-hydrogen. At ambient temperature, approximately 75% of hydrogen is in the ortho form. If liquefied without conversion, ortho-hydrogen spontaneously converts to the para form inside the storage tank, releasing heat and causing massive boil-off losses of up to 50% within a few days. This conversion is therefore forced at multiple stages of the cooling process using catalysts such as iron oxide or chromium oxide. The fourth and final stage is the Joule-Thomson expansion. Pressure is suddenly reduced through a throttling valve or expansion turbine, causing the temperature to drop to 20 K and the hydrogen to transition to the liquid state.

5.4 On-Board Storage Challenges

The cryogenic tank installed on board a truck is a Dewar vessel, a double-walled container with a high vacuum between the walls to minimise heat transfer. Despite this insulation, a residual boil-off is unavoidable. A truck left stationary for an extended period must therefore vent excess pressure through a boil-off management system.

6 simulation on a virtual tank

6.1 Establishment of the Model

In order to evaluate the thermal behaviour of the liquid hydrogen (LH₂) tank used in the hydrogen-powered truck, a simplified but physically consistent model was established. The tank is assumed to have a capsule geometry, composed of a cylindrical section and two hemispherical end-caps. A schematic representation of the tank and the insulation layers is shown in Figure 7.

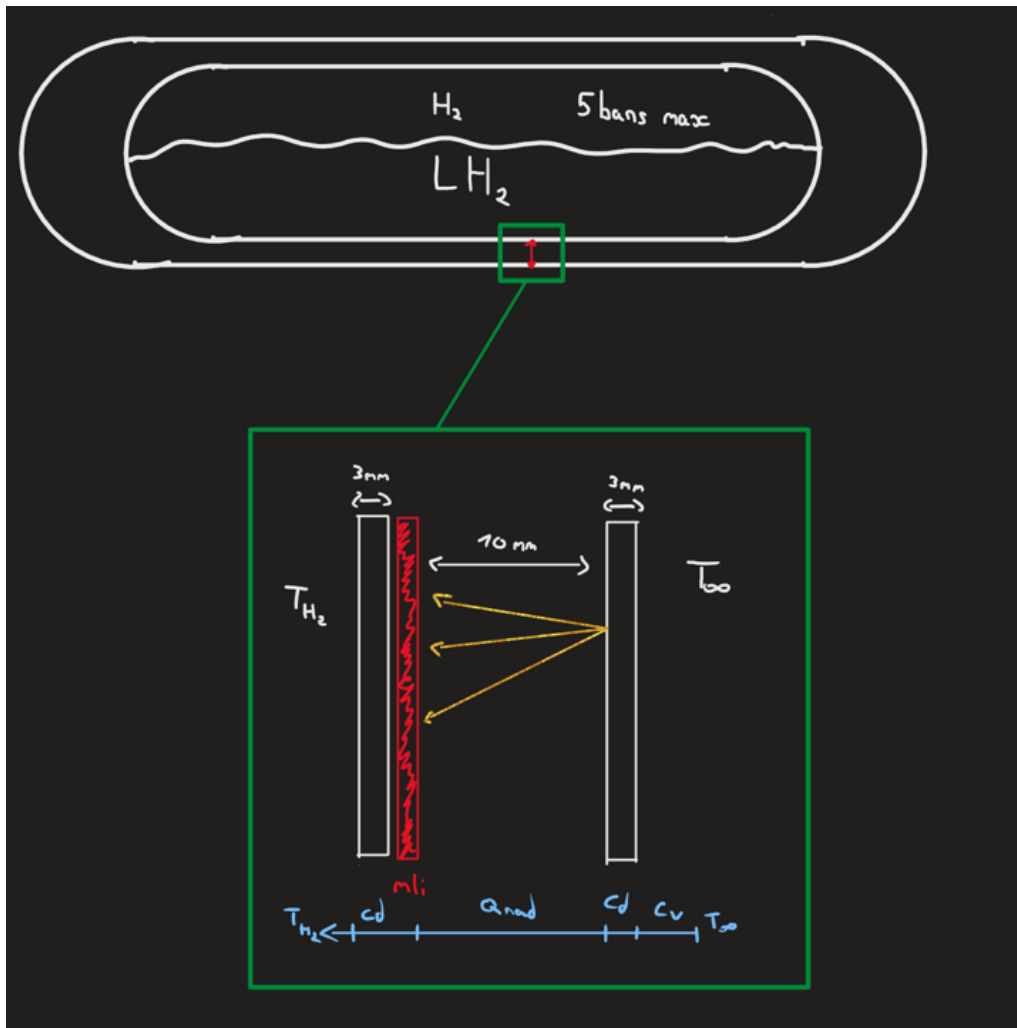


Figure 4: Schematic representation of the LH₂ tank, the stainless-steel walls, the vacuum gap and the MLI insulation.

Material properties

- Stainless-steel wall thickness: $e_{\text{inox}} = 3 \text{ mm}$.
- Stainless-steel thermal conductivity: $k_{\text{inox}} = 15 \text{ W/m/K}$.
- Vacuum gap thickness: $e_{\text{gap}} = 2 \text{ mm}$.
- Effective thermal conductivity of the residual gas: $k_{\text{gap}} = 10^{-3} \text{ W/m/K}$.
- Effective emissivity of the MLI layer: $\varepsilon_{\text{MLI}} = 10^{-2}$.

Convective heat transfer

- Internal convection coefficient (LH₂ mixing/boiling): $h_{\text{int}} = 500 \text{ W/m}^2/\text{K}$.
- External convection coefficient (ambient air): $h_{\text{ext}} = 10 \text{ W/m}^2/\text{K}$.

Thermophysical properties of liquid hydrogen

- Boiling temperature: $T_{\text{boil}} = 20.3 \text{ K}$.
- Density: $\rho = 70 \text{ kg/m}^3$.

- Specific heat capacity: $c_p = 9700 \text{ J/kg/K}$.
- Latent heat of vaporisation: $L_v = 4.45 \times 10^5 \text{ J/kg}$.

Thermal modelling assumptions

- The LH_2 is considered as a single, well-mixed thermal node (uniform temperature).
- The stainless-steel wall has negligible thermal inertia compared to the liquid.
- Radiative heat transfer across the vacuum gap is reduced by the MLI and modelled using an effective emissivity.
- When the liquid reaches T_{boil} , all incoming heat is used for vaporisation.

6.2 linear study

In order to evaluate the thermal behaviour of the liquid hydrogen (LH) tank, a lumped-parameter thermal model (1D) is adopted. This approach provides a good compromise between physical realism and computational simplicity, and is commonly used in preliminary cryogenic tank studies.

To determine the equations for this first model, we start with a total energy balance, but first, let's review the 1st Thermodynamic Intern Energy Conservation Principle :

$$dU = \delta Q + \delta W$$

However, in our case, under the simplifying assumptions, we do not consider mechanical work to be significant; therefore :

$$dU = \delta Q$$

In the present case, the LH_2 tank is considered as a closed system with no mechanical work exchange and negligible volume variation. Therefore, the energy balance reduces to a purely thermal form. Assuming a homogeneous liquid phase, the internal energy can be expressed as:

$$U = mc_p T$$

;

Differentiating with respect to time yields:

$$mc_p \frac{dT}{dt} = \phi_{\text{tot}}$$

The total heat flux entering the tank originates from several physical mechanisms acting in series between the ambient air and the liquid hydrogen.

These include:

- External Convection from ambient air to the outer tank wall
- Conduction through the external steel wall

- Radiative heat transfer across the MLI
- Conduction through the internal steel wall

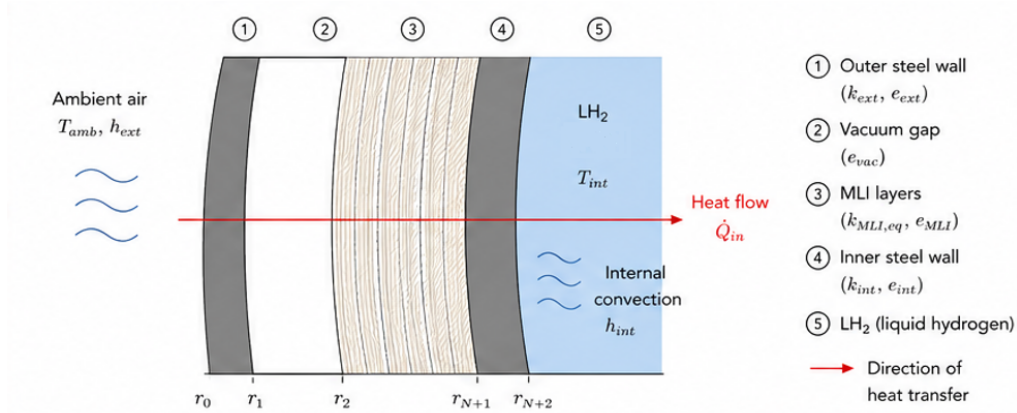


Figure 5: Zoom on the LH₂ tank, the stainless-steel walls, the vacuum gap and the MLI insulation.

These different contributions can be conveniently combined using a thermal resistance network analogy, leading to the following expression:

$$\phi_{tot} = \frac{T_{\infty} - T_{LH_2}}{R_{tot}}$$

Where: R_{tot} represents the sum (in series) of all thermal resistances between the ambient environment and the liquid hydrogen. This formulation provides a convenient and physically meaningful way to aggregate multiple heat transfer mechanisms into a single expression.

6.2.1 Detailed heat transfer mechanisms

Convection

Computed using the convective heat transfer coefficient h_{ext} and surface area A :

$$R_{convec} = \frac{1}{h_{ext}A}$$

Conduction

The conductive heat transfer through the tank wall is described by Fourier's law and computed using the Stainless-steel thermal conductivity k_{inox} and its thickness e_{inox} :

$$R_{conduct} = \frac{e_{inox}}{k_{inox}A}$$

Radiative heat transfer

It is modelled using an effective emissivity approach derived from the Stefan–Boltzmann law after linearisation:

$$R_{rad} = \frac{1}{\epsilon_{MLI}\sigma A(T_{ext}^2 + T)(T_{ext} + T)}$$

Where: $\sigma = 5.67 \times 10^{-8} \text{ W.m}^{-2}\text{K}^{-4}$

This simplified model provides a physically grounded and computationally efficient framework to analyse the thermal behaviour of cryogenic hydrogen storage systems. While it neglects spatial temperature gradients and some complex phenomena, it captures the dominant heat transfer mechanisms and accurately predicts both temperature evolution and evaporation dynamics. Such an approach is particularly suitable for preliminary design studies and system-level performance evaluation.

6.3 Surface Study

This subsection presents the numerical simulation performed to evaluate the thermal evolution of the LH_2 inside the tank. The model includes conduction through the stainless-steel wall, conduction through the vacuum gap, radiative heat transfer through the MLI, internal and external convection, and the phase change of hydrogen at its boiling temperature.

Geometrical assumptions

- Internal radius of the tank: $R = 0.5 \text{ m}$.
- Cylindrical length: $L_{\text{cyl}} = 3.0 \text{ m}$.
- Internal volume:

$$V = \pi R^2 L_{\text{cyl}} + \frac{4}{3} \pi R^3.$$

- Internal heat exchange surface:

$$A = 2\pi R L_{\text{cyl}} + 4\pi R^2.$$

The simulation is carried out over two different time windows:

- the first hour, to observe the detailed temperature evolution,
- the full duration until complete evaporation of the liquid hydrogen.

Python implementation

Now that the model is properly introduced, we can simulate the behavior of the liquid hydrogen inside the tank to see how it handles the environment.

Analysis of the Results

The first plot shows that, despite the different insulation strategies implemented around the tank including the stainless-steel double wall, the vacuum gap and the multilayer insulation the liquid hydrogen still reaches its boiling temperature extremely quickly. The temperature rises from 20.0 K to 20.3 K in approximately 6 minutes, which highlights the strong thermal driving force imposed by the ambient environment. Even with efficient insulation, the combination of conductive, convective and radiative heat transfer inevitably leads the LH_2 to its saturation temperature in a short period of time. This behaviour confirms that the initial thermal transient is dominated by the unavoidable heat leak through the tank structure, and that no insulation system can prevent the liquid from eventually reaching its boiling point.

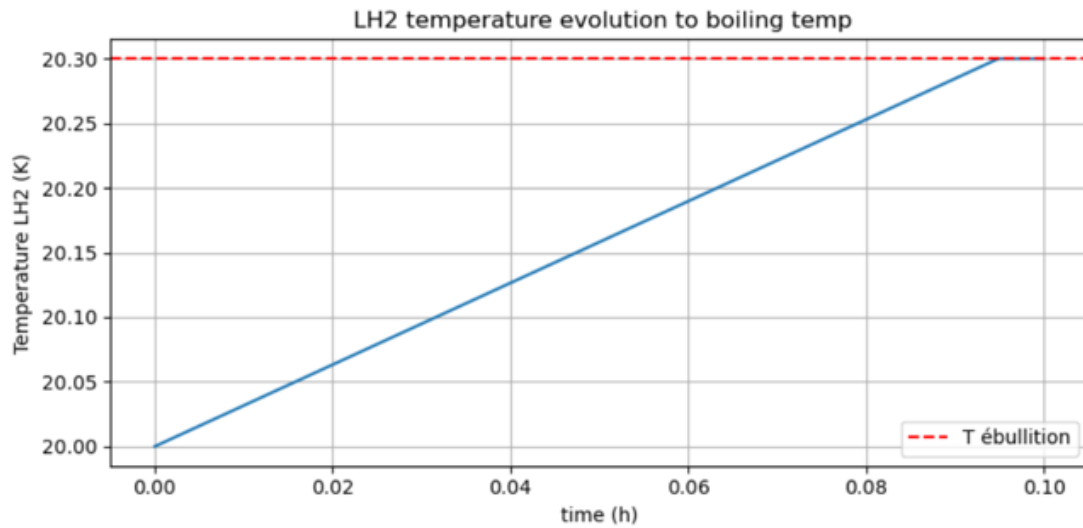


Figure 6: “Evolution of the liquid hydrogen temperature during the initial heating phase, showing the rise from 20.0 K up to the boiling point at 20.3 K.”

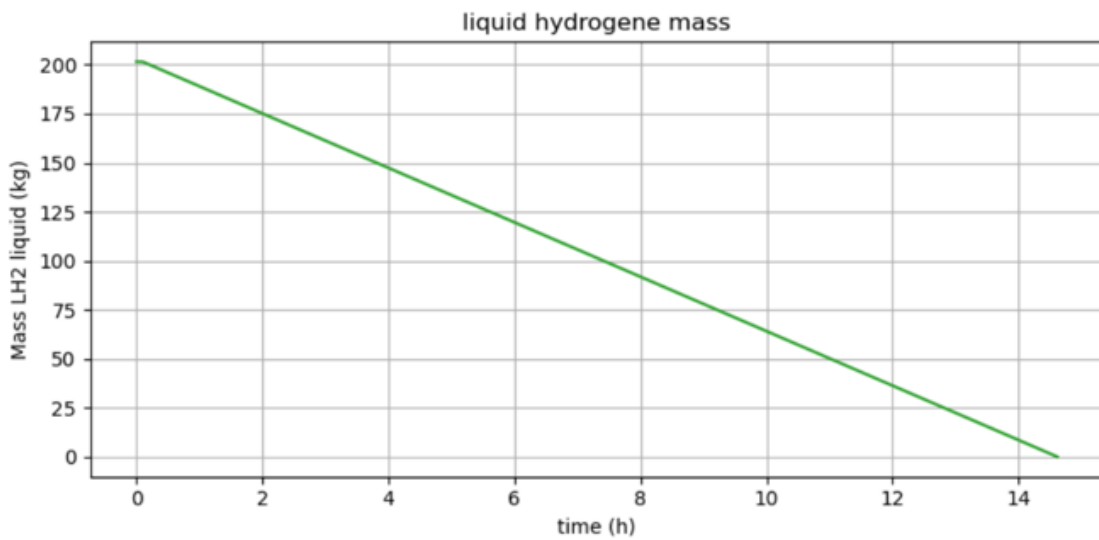


Figure 7: “Time evolution of the remaining liquid hydrogen mass inside the tank, illustrating the complete evaporation process under the imposed thermal loads.”

For this reason, the second plot focuses on the evolution of the remaining liquid hydrogen mass once the boiling temperature has been reached. At this stage, all incoming heat is converted into latent heat, leading to a continuous and nearly linear vaporisation of the LH₂. The mass decreases steadily until complete evaporation, providing a direct measure of the tank’s cryogenic endurance under the imposed thermal loads. This long-term behaviour shows us that the mass of liquid hydrogen, if let untouched, will completely vaporize in 14.6 hours which begs some issues as for its concrete use for trucks.

7 Discussion and Conclusion

7.1 Weaknesses of Liquid Hydrogen Storage

Despite the promising use of liquid hydrogen in storage of providing stations, the thermal study conducted on the capsule-shaped cryogenic tank highlights several intrinsic limitations associated with the use of liquid hydrogen in mobile applications. Despite the presence of a stainless-steel double

wall, a vacuum gap and multilayer insulation, the liquid hydrogen reaches its boiling temperature extremely rapidly due to the large temperature gradient between the cryogenic fluid and the ambient environment. Once the boiling point is reached, the tank continuously produces gaseous hydrogen through vaporisation. Because the internal pressure must remain below the safety limit of 5 bar, the evaporated hydrogen cannot be stored indefinitely and must be vented to the atmosphere. This phenomenon, known as boil-off, leads to a permanent loss of usable fuel and reduces the effective autonomy of the vehicle.

In addition to material losses, liquid hydrogen storage introduces significant safety challenges. The combination of cryogenic temperatures, pressure build-up and mandatory venting requires complex regulation systems and strict operational procedures. Any malfunction in the insulation, venting or pressure-control system may result in rapid pressure rise, structural stress on the tank or uncontrolled hydrogen release. These constraints make liquid hydrogen difficult to manage in applications where long storage times, frequent stops or variable environmental conditions are expected. The results of our simulations clearly illustrate these weaknesses: the liquid reaches its boiling point within minutes, and the subsequent evaporation proceeds steadily until complete depletion of the stored hydrogen.

7.2 Supervisor Recommendations and Orientation Toward Compressed Hydrogen

Following the analysis of these results, our project supervisor encouraged us to reconsider the suitability of liquid hydrogen for the intended application. The rapid onset of boil-off, the continuous loss of material and the associated safety constraints make cryogenic storage poorly adapted to the operational profile of a hydrogen-powered truck. Instead, the supervisor recommended exploring the use of compressed gaseous hydrogen at high pressure, particularly around 500 bar.

High-pressure gaseous storage avoids the fundamental limitations of cryogenic systems. It does not suffer from boil-off losses, does not require extreme insulation, and eliminates the need for venting. Although the volumetric energy density of gaseous hydrogen is lower than that of liquid hydrogen, the absence of continuous mass loss and the simplicity of the storage system make it far more robust for mobile applications. This perspective aligns with current industrial trends, as the hydrogen mobility sector increasingly favours 350 bar and 700 bar compressed storage solutions for heavy-duty vehicles, while liquid hydrogen remains mostly restricted to aerospace and specialised cryogenic applications.

7.3 Comparison Between Liquid Hydrogen and 500 bar Gaseous Hydrogen

Table 4: Comparison between liquid hydrogen and 500 bar gaseous hydrogen for mobility applications.

Criterion	Liquid Hydrogen (LH ₂)	Gaseous Hydrogen at 500 bar
Storage temperature	20 K (cryogenic)	Ambient temperature
Pressure management	Must remain < 5 bar; requires venting	Stable at 500 bar; no venting required
Material losses	Continuous boil-off; unavoidable mass loss	No boil-off; no continuous losses
Insulation requirements	Complex: vacuum gap + MLI + double wall	Minimal insulation required
Safety considerations	Cryogenic risks, pressure rise, venting hazards	High-pressure risks but well-controlled industrially
System complexity	High (cryogenics, venting, thermal management)	Moderate (pressure vessels, regulators)
Industrial adoption	Limited to aerospace and specialised uses	Widely adopted for mobility (350/700 bar)
Suitability for trucks	Poor due to boil-off and safety constraints	Good: stable, robust, no mass loss

This comparison clearly shows that while liquid hydrogen offers high gravimetric energy density, its operational drawbacks make it unsuitable for the type of application considered. Compressed gaseous hydrogen at 500 bar emerges as a more practical and reliable solution, consistent with current industrial practices and the recommendations of our project supervisor.

7.4 Acknowledgments

We would finally like to express our sincere gratitude to our supervisor M. del Rosal Egido for his invaluable guidance, insightful advice and support throughout the development of this project.

His expertise and critical perspective have significantly contributed to the depth, rigor, and overall quality of our work. We are particularly grateful for the time and dedication he devoted to assisting us, as well as for his availability.

His support has been instrumental in shaping both our understanding of the subject and the successful completion of this report.

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